

Dividend Models

Valuation of Firm

- Value of the firm can be maximized if shareholders wealth is maximized.
- Accordingly

Irrelevance concept of dividend

- When dividend decision does not affect the shareholders' wealth, dividend payments hence does not affect the valuation of the firm.

Relevance concept of dividend

- When dividend decision materially affect the shareholders' wealth, dividend payments also influence valuation of the firm.

Irrelevance Theory

- Residual Theory
 - Dividend decision has no effect on the wealth of the shareholders or prices of the shares and hence it is irrelevant so far as the valuation of the firm is concerned.
 - Dividend decision is merely a part of financing decision because the earnings available may be retained in the business for re-investment.
 - If the funds are not required in the business, they may be distributed as dividend.
 - Thus, the decision to pay dividends or retain the earnings may be taken as residual decision.

Irrelevance Theory

- Modigliani Miller Theory

- Dividend policy has no effect on the market price of the share and the value of the firm is determined by the earning capacity of the firm or its investment policy.
- MM Model argues that whatever increase in the value of the firm results from the payment of dividend, will be exactly offset by the decline in the market price of shares because of external financing and there will be no change in the total wealth of the shareholders.
- If a firm, having investment opportunities, distributes all its earning among the shareholders, it will have to raise additional funds from external sources. This will result in increase in number of shares or payment of interest charges resulting in fall in the earning per share in the future.
- Whatever a shareholder gains on account of dividend payment is neutralized completely by the fall in market price of the share.

Proposition

- Since investors do not need dividends to convert shares to cash; they will not pay higher prices for firms with higher dividends.
- Investors can create whatever income stream they prefer by using homemade dividends.
- Investors can create their own dividends. Consequently, firm value will not be affected by dividend payments.

MM Model Assumptions

- **Capital markets are perfect**
- **Investors behave rationally**
- **Information on the firm is available to all without any cost**
- **No floatation cost or transaction cost associated with dividend or share**
- **No investor is large enough to influence the market price of the share**
- **There are either no taxes or there are no differences in the tax rates applicable to dividends and capital gains**
- **Firm has a rigid investment policy**
- **There is no risk or uncertainty (MM dropped this assumption later)**

Argument of MM Model: 1

- Market price of share in the beginning of a period is equal to the present value of the dividends paid at the end of the period plus the market price of the share at the end of the period.

$$P_0 = \frac{D_1 + P_1}{1 + K_e}$$

- P_0 = market price of share at beginning of the period.
- D_1 = dividend to be received at the end of the period.
- P_1 = market price per shares at the end of the period.
- K_e = cost of equity capital.

value of P_1 can be derived as:

$$P_1 = P_0 (1 + K_e) - D_1$$

Argument of MM Model: 2

- Assuming that investment required by the firm on account of payment of dividend is financed out of the new issue of equity shares.
- Number of shares can be computed as:

$$m = \frac{I - (E - nD1)}{P1} ; m = \frac{nD1}{P1} ; nD1 = mP1$$

- Value of the firm can be ascertained as:

$$nP_o = \frac{\overset{\text{New Issue}}{\uparrow} (n+m)P1 + \overset{\text{Dividend paid}}{\uparrow} (I - E)}{1+K_e} \quad P_o = \frac{D1 + P1}{1 + K_e}$$

- m = number of shares to be issued;
 - E = total earnings of the firm during the period;
 - n = no. of shares outstanding at beginning ;
 - $D1$ = dividend per share to be paid at the end of the period;
 - $P1$ = market price per share at the end of the period.
- I = investment requirement;
 K_e = cost of equity capital;
 P_o = value of the firm.

Dividend Irrelevance

- Value of Firm = Value of Equity = discounted value of future cashflows available to equity holders = discounted value of dividends (if all available cashflow is paid out).
- If all cash flows are not reinvested is paid out as dividends.

$$V_0 = \sum_{t=0}^{\infty} \frac{D_t}{(1 + k_e)^t}$$

$$V_0 = \sum_{t=0}^{\infty} \frac{NCF_t - I_t}{(1 + k_e)^t}$$

Dividend Irrelevance

- MM used a **source and application of funds** argument to show that Dividend Policy is irrelevant.
- Source of Funds = Application of Funds

$$NCF_t + NS_t = I_t + Div_t$$

$$\Rightarrow NCF_t - I_t = Div_t - NS_t$$

$$V_0 = \sum_{t=0}^{\infty} \frac{D_t - NS_t}{(1 + k_e)^t} = V_0 = \sum_{t=0}^{\infty} \frac{NCF_t - I_t}{(1 + k_e)^t}$$

Dividend Irrelevance

- Dividends do not appear in the equation.
- If the firm pays out too much dividend, it issues new equity to be able to reinvest. If it pays out too little dividend, it can use the balance to repurchase shares.
- Assumption is the availability of finance in the capital market.

$$V_0 = \sum_{t=0}^{\infty} \frac{NCF_t - I_t}{(1 + k_e)^t}$$

Earnings Invested (No Dividend)

- New Investment Requirement 1000; $K_e = 20\%$

Cash	1000	Debt	0
Assets	9000	Equity (10 x 1000)	10000
		New issue	
Total	10000		10000
Market price	60		

Cash	2000	Debt	0
Assets	9000	Equity (10 x 1000)	10000
Investment	1000	Earnings	2000
Total	12000		12000
Market price	60		

Investment through New Issue (Dividend Paid)

- New Investment Requirement 1000; $K_e = 20\%$
- Dividend paid = $1000/1000 = 1$ per share

Cash		Debt	0
		Equity (9 x 1000)	9000
Assets	9000		
Investment / Dividend (1000)	1000	New issue (17 x 59)	1000
Total	10000		10000
Market price	59		

		Equity (9 x 1000)	9000
Cash	2000		
		New issue (17 x 59)	1000
Assets	9000		
Investment / Dividend (1000)	1000	Earnings	2000
Total	12000		12000
Market price	59		

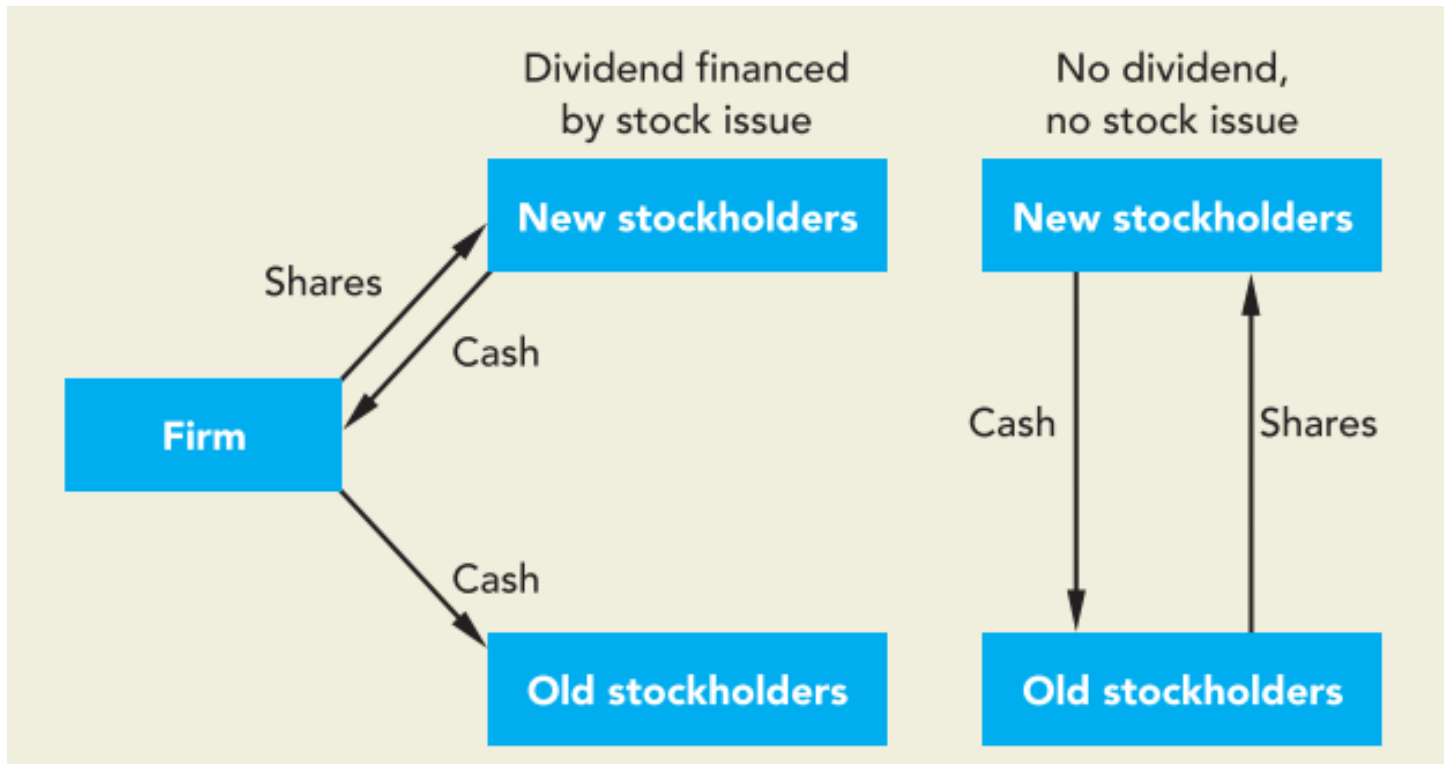
- Market Price = $60 - 1 = 59$; New Shares = $(1000 \times 1)/59 = 17$ shares

Home Made Dividend

- Market Price 60; Dividend per share 1; Shares held by investor 100
- Investor needs a dividend of 6.9 per share

Dividend per share	1	6.9
	Home made Dividend	Dividend
Cash from dividend	100	690
Cash from sale of 10 shares (ex-dividend)	590	
Total Cash	690	690
Value of shares	5310	5310
Value of Share	(59 x 90)	(53.1 x 100)
Market Price (Ex-dividend)	(60-1) 59	(60-6.9) 53.1

Dividend Vs. No Dividend



Source: Corporate Finance, Breley and Meyers.

Relevance Theory

- Dividends communicate information to the investors about the firm's profitability and hence dividend decisions becomes relevant.
- Those firms which pay higher dividends, will have greater value as compared to those which do not pay dividends or have a lower dividend pay out ratio.
- James Walter
- Myron Gordon
- Jone Linter
- Richardson

Walter's Dividend Model

- Relationship between the internal rate of return (r) earned by the firm and cost of capital (k) is very significant in determining the dividend policy to maximize the wealth of the shareholders.
- If $r > k$, the firm is earning higher rate of return, the firm should retain the earnings. Such firms are termed as growth firms and the optimum pay out would be zero.
- If $r < k$, the optimum payout would be 100% and the firm should distribute the entire earning as dividend.
- If $r = k$, the dividend policy will not affect the market value of shares as the shareholders will get the same returns from the firm as expected by them.

Assumptions of Walter's Model

- Investment of the firm are financed through retained earnings only and the firm does not use external source of fund.
- Internal rate of return (r) and the cost of capital (k) of the firm are constant.
- Earnings and dividends do not change while determining the value.
- Firm has a long life.

Value of the Firm $P = \frac{D}{K_e - g}$.

P = price of equity shares.

D = initial dividend per share.

K_e = cost of equity capital.

g = expected growth rate of earnings.

Value of the Firm (Walter's Model)

$$P = \frac{D}{K_e} + \frac{r(E - D)}{K_e}$$

- P = market price per share.
- D = dividend per share.
- r = internal rate of return.
- E = earning per share.
- K_e = cost of equity capital.

Example

- $r > k_e$ Dividend = 0; Max Price = 200
- $r < k_e$ Dividend = Earnings; Max Price = 100

	$r = 20\% \quad k_e = 10\%$			$r = 5\% \quad k_e = 10\%$			$r = 10\% \quad k_e = 10\%$	
	$r > k_e$			$r < k_e$			$r = k_e$	
Price	Earnings	Dividend	Price	Earnings	Dividend	Price	Earnings	Dividend
200	10	0	50	10	0	100	10	0
180	10	2	60	10	2	100	10	2
160	10	4	70	10	4	100	10	4
140	10	6	80	10	6	100	10	6
120	10	8	90	10	8	100	10	8
100	10	10	100	10	10	100	10	10

Criticism of Walter's Model

- Firms raise funds by external financing.
- Internal rate of return does not remain constant.
- Assumption that cost of capital will remain constant does not hold good.
 - Firm's risk pattern does not remain constant, hence k_e will not remain constant.

Gordon's Model

- Growth rate of the firm determines the valuation of the firm in the market
- Growth rate is influenced by the retention rate of the firm, where retention rate is defined as $1 - \text{dividend payout rate}$.
- When a firm utilizes all its earnings for dividend payment, the retention rate being zero signifies that the firm does not have investment projects greater than the cost of capital.
- Alternatively, a zero-dividend payment implies a 100% retention rate where the firm is assumed to utilize all its earnings in profitable investment projects giving returns higher than the cost of capital.

Assumption of Gordon's Model

- Firm is an all-equity firm.
- No external financing is available or used. Retained earnings represent the only source of financing investment programs.
- Rate of return on the firm's investment is constant.
- Retention ratio, b , once decided upon is constant thus the growth rate of the firm $g = br$, is also constant.
- Cost of capital of the firm remains constant and it is greater than the growth rate, $K > br$.
- Corporate taxes do not exist.

Value of Firm (Gordon's Model)

$$P = \frac{E(1 - b)}{K_e - br} \quad P_0 = \frac{D_1}{K_e - g} = \frac{D_0(1 + g)}{K_e - g}$$

- P = price of share
- E = earning per share
- b = retention ratios
- K_e = cost of equity capital
- br = g = growth rate in r
- D₀ = dividend per share
- D₁ = expected dividend at the end of year 1

Example

- Dividend = 0; Max Price = 0
- Dividend = Earnings; Max Price = 210

	g = 5% ke=10%	
Price	Earnings	Dividend
21	10	1
42	10	2
84	10	4
126	10	6
168	10	8
210	10	10

Finite Supernormal Profit Model

- Rate of return on Investment > market required return for T years.
- After that, Rate of Return on Investment = Market required return.

$$P_0 = \frac{NCF_1}{k_e} + \frac{RR \times NCF_1 \times T \times (r - k_e)}{k_e(1 + k_e)}$$

- Net Cash Flow (NCF1) = 100; $k_e = 10\%$; $T = 5$ years; Retention rate $RR = 0.1$
- Return (r) = 20% for 10 years and thereafter 10%
- $P_0 = (100 / .1) + (0.1 \times 100 \times 5 \times [(0.20 - 0.1) / \{0.1(1 + 0.1)\}]) = 105$

Gordon's Revised Model

- Assumption in Gordon's basic valuation model that cost of capital (k) remains constant for a firm is not true in practice .
- Gordon revised this basic model to consider risk and uncertainty.
- Revised model suggested that even when $r = k$, dividend policy affects the value of shares.
- On account of uncertainty of future, shareholders discount future dividends at higher rate than they discount near dividends.

Lintner's Model

- Lintner model is an economic formula for determining the optimal dividend policy for a firm.
- Firm can evaluate the effectiveness of its dividend policy.
- Model considers the target payout ratio and the speed at which current dividends adjust to the target.
- Firm's current year's dividend is dependent on its current year's earnings and last year's dividend.
- Dividends are thus, defined as the weighted average of a firm's past earnings.
- Model theorizes the process in which a publicly-traded firm sets its dividend policy.
- It logically assumes that every firm maintains a constant rate of dividend even if the results in a particular period are not up to the benchmark.
- It assumes that investors will prefer to receive a particular dividend payout.

Lintner's Model Assumptions

- Firm will have to change its dividend whenever there is a change in its earnings if the firm is consistent with its target payout.
- Model suggests that there are two parameters to be considered in dividend policy.
 - Payout ratio
 - Speed at which current dividends adjust to its target.
- Firms set long-run target dividend payout ratios.
- Firm's managers are more interested in change in the dividend rather than in dividend payout.
- There is a tendency for dividends to follow earnings, but they rather follow a smoother path than earnings
- Dividends are sticky in nature because managers are unwilling to effect dividend changes.

Dividend Payout of a Firm (Lintner's Model)

- $D_t = D_y + af \times [(E_t \times tp) - D_y]$
- where:
- $y = t - 1$
- D_t = Dividend per share at time ($t = 1$).
- D_y = Dividend per share at time ($t-1$), (last year's dividend per share)
- af = Speed of adjustment rate or the partial adjustment coefficient (with 0 less than or equal to af and af less than or equal to 1).
- E_t = Earnings per share (or free-cash-flow per share) at time t .
- tp = Target payout ratio on earnings per share (free-cash-flow per share) (with 0 less than or equal to tp and tp less than or equal to 1).

Example

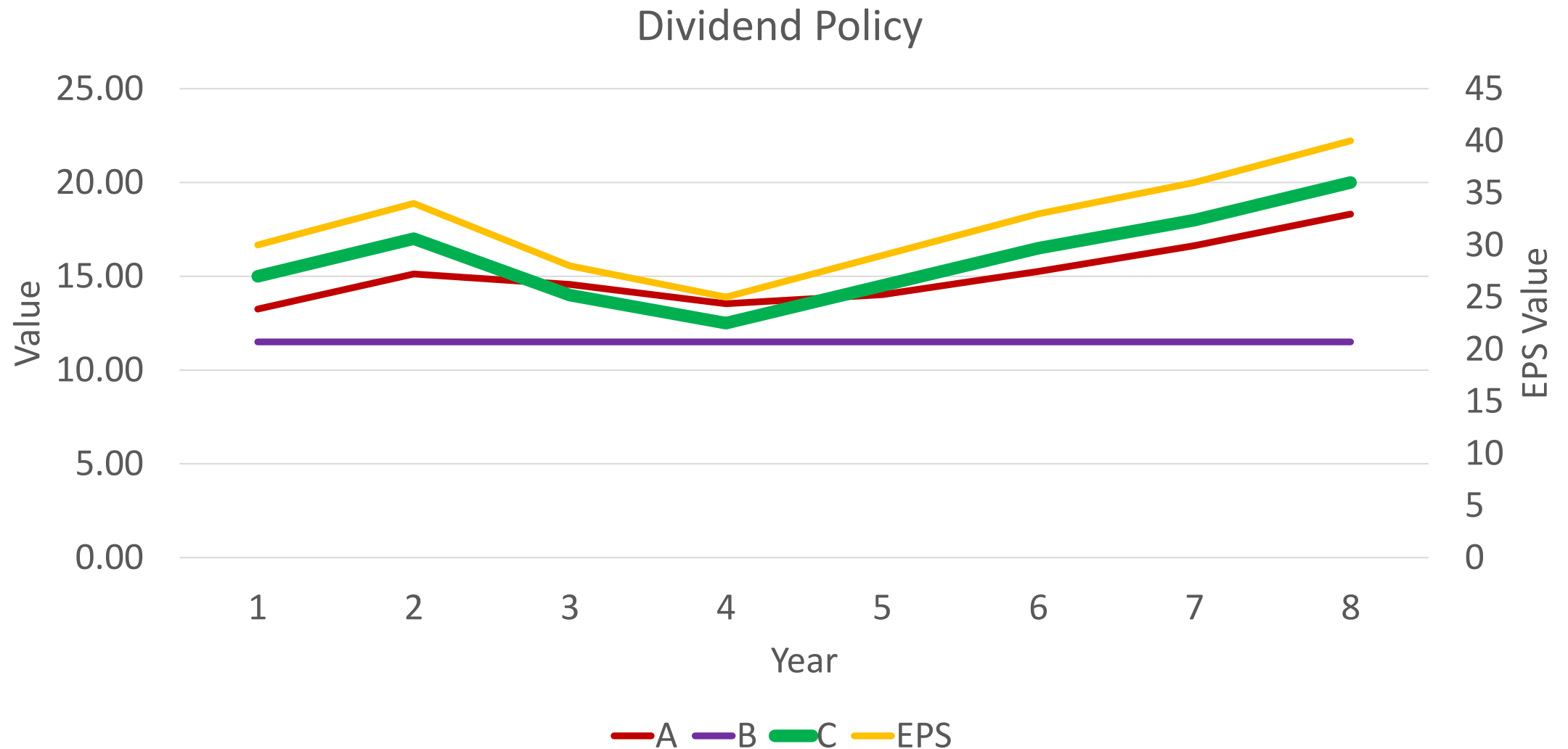
- $D_t = D_y + a_f \times [(E_t \times t_p) D_y]$: Target Payout Ratio (t_p) = 0.5

	Firm	A	B	C
	a_f	0.5	0	1
Year	EPS	Dividend	Dividend	Dividend
1	30	13.25	11.50	15.00
2	34	15.13	11.50	17.00
3	28	14.56	11.50	14.00
4	25	13.53	11.50	12.50
5	29	14.02	11.50	14.50
6	33	15.26	11.50	16.50
7	36	16.63	11.50	18.00
8	40	18.31	11.50	20.00

-----Previous year

Assumption

Dividend Policy of Firms



Implications of Lintner's Model

- No Dividends is a viable dividend policy: Firm will not distribute any of its earnings as dividends to its shareholders. Firm's management will rather allocate any extra earnings towards expansion and future growth.
- Stable Dividend Policy: Firm shares its dividends at a given rate to its shareholders at set intervals. Quarterly dividend distributions are the most common. This policy can be implemented in various ways.
 - Constant payout ratio dividend.
 - Constant dividend value.
 - A combination of the first two policies.
- Irregular Dividends Policy: Mostly used when a firm lacks a steady inflow of liquid funds or its earnings fluctuations too much.

Applying Lintner's Model to Firms

- Firm data on Earnings Per Share and Dividend for ten years can be regressed as:

$$Div_t = \alpha + (1 - af)Div_{t-1} + af \times tp \times Eps_t$$

$$Div_t = a + bDiv_{t-1} + cEps_t$$

- Parameters give us the following information:

$$a = 0, af = 1 - b, tp = c / (1 - b).$$

Business Scenario

- Assume an unlevered furniture manufacturing firm, has operating income after taxes of INR 100 million, is growing at 5 percent a year, and has a cost of capital of 10 percent. Furthermore, assume that this firm has reinvestment needs of INR 50 million, also growing at 5 percent a year, and there are 105 million shares outstanding. Assume that this firm pays out residual cash flows as dividends each year.
- Prove the dividend irrelevance theory.

Case Information

- Free Cash Flow to the Firm = $\text{EBIT} (1 - \text{Tax Rate}) - \text{Reinvestment Needs} = \text{INR } 100 \text{ million} - \text{INR } 50 \text{ million} = \text{INR } 50 \text{ million}$
- Value of the Firm = $\text{FCFF} (1 + g) / (\text{Cost of capital} - g) = 50(1.05) / (0.10 - 0.05) = \text{INR } 1,050 \text{ million}$
- Price per Share = $\text{INR } 1,050 \text{ million} / 105 \text{ million} = \text{INR } 10.00$
- Based on its cash flows, this firm could pay out INR 50 million in dividends.
- Dividend per Share = $\text{INR } 50 \text{ million} / 105 \text{ million} = \text{INR } 0.476$
- Total Value per Share = $\text{INR } 10.00 + \text{INR } 0.476 = \text{INR } 10.476$
 - The total value per share measures what shareholders get in price and dividends from their stock holdings.

Scenario1 (Firm doubles dividends)

- Firm pays out INR 100 million in dividends instead of INR 50 million.
- It now has to raise INR 50 million in new financing to cover its reinvestment needs.
- Assume the firm can issue new stock with no issuance cost to raise these funds. If it does so, the firm value will remain unchanged, because the value is determined not by the dividend paid but by the cash flows generated on the projects. Growth rate and the cost of capital are unaffected.
- Value of the Firm = $FCFF(1+g)/(Cost\ of\ capital - g) = 50(1.05)/(0.10 - 0.05) = \text{INR } 1,050\ \text{million}$
- Existing shareholders will receive a much larger dividend per share, because dividends have been doubled:
- Dividends per share = $\text{INR } 100\ \text{million} / 105\ \text{million shares} = \text{INR } 0.953$
- Price per share at which the new stock will be issued: (after the new stock issue of INR 50 million, the old shareholders in the firm will own only INR 1,000 million of the total firm value of INR 1,050 million).
- Value of the firm for existing shareholders after dividend payment = INR 1,000 million Price per Share = $\text{INR } 1,000\ \text{million} / 105\ \text{million} = \text{INR } 9.523$
- Price per share is now lower than it was before the dividend increase, but it is offset by the increase in dividends.
- Value Accruing to Shareholder = $\text{INR } 9.523 + \text{INR } 0.953 = \text{INR } 10.47$

Scenario 2: (Firm does not pay dividend)

- Firm retains the residual INR 50 million as a cash balance.
- Value of the firm to existing shareholders:
- Value of Firm = Present Value of After-Tax Operating CF + Cash Balance = $50(1.05) / (0.10 - 0.05) + \text{INR } 50 \text{ million} = \text{INR } 1,100 \text{ million}$
- Value per Share = $\text{INR } 1,100 \text{ million} / 105 \text{ million shares} = \text{INR } 10.476$

Dividend Irrelevance

- Value to Shareholders

Value of Firm : Operating cash flows	Dividends	Value to existing shareholders	Price per share	Dividend per share	Total value per share
1,050	0	1,100	10.48	0	10.48
1,050	10	1,090	10.38	0.10	10.48
1,050	20	1,080	10.29	0.19	10.48
1,050	30	1,070	10.19	0.29	10.48
1,050	40	1,060	10.10	0.38	10.48
1,050	50	1,050	10	0.48	10.48
1,050	60	1,040	9.90	0.57	10.48
1,050	70	1,030	9.81	0.67	10.48
1,050	80	1,020	9.71	0.76	10.48
1,050	90	1,010	9.62	0.86	10.48
1,050	100	1,000	9.52	0.95	10.48